

Liénard–Wiechert Potential and the Field Tensor

Cambridge Electromagnetism, Chapter 7 exercises, Question 6; accompanies David Tong's Electromagnetism.

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Electrodynamics

For a relativistic particle of charge q on a trajectory $y^\mu(\tau)$, where τ is proper time, the current density 4-vector is

$$J^\mu(x) = qc \int d\tau \delta^{(4)}(x - y(\tau)) \dot{y}^\mu(\tau), \quad (1)$$

with $\dot{y}^\mu \dot{y}_\mu = -c^2$ and $\dot{y}^0 > 0$. Show that the 4-vector potential is given by

$$A^\mu(x) \quad (2)$$

$$\begin{aligned} &= \frac{\mu_0}{2\pi} \int d^4z \Theta(x^0 - z^0) \delta(\eta_{\alpha\beta}(x^\alpha - z^\alpha)(x^\beta - z^\beta)) J^\mu(z) \\ &= -\frac{\mu_0 qc}{4\pi} \frac{\dot{y}^\mu(\tau^*)}{R^\nu(\tau^*) \dot{y}_\nu(\tau^*)}, \end{aligned} \quad (3)$$

where $R^\nu(\tau) = x^\nu - y^\nu(\tau)$ and τ^* is determined by $R^\mu(\tau^*) R_\mu(\tau^*) = 0$ and $R^0(\tau^*) > 0$.

Verify that the Lorenz gauge condition $\partial_\mu A^\mu = 0$ holds and show that

$$F^{\mu\nu} = -\frac{\mu_0 qc}{4\pi} \frac{1}{(R^\rho \dot{y}_\rho)^2} (R^\mu S^\nu - R^\nu S^\mu), \quad (4)$$

where

$$S^\nu = \ddot{y}^\nu - \frac{\dot{y}^\nu}{R^\rho \dot{y}_\rho} (c^2 + R^\tau \ddot{y}_\tau) \quad (5)$$

and all quantities on the right are evaluated at τ_* . Check this result for the case of a stationary charge at the origin.

Solution Given the 4-potential satisfies

$$\square A^\mu(x) = -\mu_0 J^\mu(x). \quad (6)$$

To solve this, we shall first find the retarded Green's function of the d'Alembert equation, which satisfies

$$\square G^{\mu\nu}(x, z) = \delta^{(4)}(x - z) \eta^{\mu\nu}. \quad (7)$$

In the Fourier domain, we have

$$-k^2 \tilde{G}^{\mu\nu}(k) = \eta^{\mu\nu}. \quad (8)$$

Implementing the retarded prescription, we have

$$G^{\mu\nu}(x, 0) = -\eta^{\mu\nu} \int \frac{d^3\mathbf{k}}{(2\pi)^3} \int \frac{dk^0}{2\pi} \frac{e^{ik^0 x^0 - i\mathbf{k}\cdot\mathbf{x}}}{(k^0 - i\epsilon)^2 - |\mathbf{k}|^2} \quad (9)$$

$$= -\eta^{\mu\nu} \Theta(x^0) \int \frac{d^3\mathbf{k}}{(2\pi)^3 \cdot 2|\mathbf{k}|} i e^{-i\mathbf{k}\cdot\mathbf{x}} [e^{i|\mathbf{k}|x^0} - e^{-i|\mathbf{k}|x^0}] \quad (10)$$

$$= \eta^{\mu\nu} \Theta(x^0) \int \frac{d^3\mathbf{k}}{(2\pi)^3 |\mathbf{k}|} e^{-i\mathbf{k}\cdot\mathbf{x}} \sin(|\mathbf{k}|x^0) \quad (11)$$

$$= \frac{\eta^{\mu\nu}}{2\pi^2 |\mathbf{x}|} \Theta(x^0) \int_0^\infty d|\mathbf{k}| \sin(|\mathbf{k}|x^0) \sin(|\mathbf{k}||\mathbf{x}|) \quad (12)$$

$$= \frac{\eta^{\mu\nu}}{4\pi^2 |\mathbf{x}|} \Theta(x^0) \int_0^\infty d|\mathbf{k}| [\cos(|\mathbf{k}|(x^0 - |\mathbf{x}|)) - \cos(|\mathbf{k}|(x^0 + |\mathbf{x}|))] \quad (13)$$

$$= \frac{\eta^{\mu\nu}}{4\pi^2 |\mathbf{x}|} \Theta(x^0) \cdot \pi [\delta(x^0 - |\mathbf{x}|) - \delta(x^0 + |\mathbf{x}|)] \quad (14)$$

$$= \frac{\eta^{\mu\nu}}{4\pi |\mathbf{x}|} \Theta(x^0) \delta(x^0 - |\mathbf{x}|), \quad (15)$$

where we used $\int_0^\infty d|\mathbf{k}| \cos(|\mathbf{k}|a) = \pi \delta(a)$, and the second delta function vanishes under $\Theta(x^0)$ since $x^0 + |\mathbf{x}| > 0$ always. Using $\Theta(x^0) \delta(x^2) = \Theta(x^0) \delta(x^0 - |\mathbf{x}|) / (2|\mathbf{x}|)$, we obtain

$$\boxed{G_{\text{ret}}^{\mu\nu}(x, 0) = \frac{\eta^{\mu\nu}}{2\pi} \Theta(x^0) \delta(x^2).} \quad (16)$$

Therefore, the covariant retarded potential can be written as

$$A^\mu(x) = \frac{\mu_0}{2\pi} \int d^4z \Theta(x^0 - z^0) \delta(\eta_{\alpha\beta}(x^\alpha - z^\alpha)(x^\beta - z^\beta)) J^\mu(z). \quad (17)$$

Using the Dirac delta identities, we can simplify this:

$$\begin{aligned} A^\mu(x) &= \frac{\mu_0 qc}{2\pi} \int d\tau \int d^4z \Theta(x^0 - z^0) \delta(\eta_{\alpha\beta}(x^\alpha - z^\alpha)(x^\beta - z^\beta)) \delta^{(4)}(z - y(\tau)) \dot{y}^\mu(\tau) \\ &= \frac{\mu_0 qc}{2\pi} \int d\tau \Theta(x^0 - y^0(\tau)) \delta(R^\nu(\tau) R_\nu(\tau)) \dot{y}^\mu(\tau) \\ &= \frac{\mu_0 qc}{4\pi} \int d\tau \Theta(x^0 - y^0(\tau)) \sum_i \frac{\delta(\tau - \tau_i)}{|R^\nu(\tau_i) \dot{y}_\nu(\tau_i)|} \dot{y}^\mu(\tau), \end{aligned} \quad (18)$$

where τ_i satisfy $R^\nu(\tau_i) R_\nu(\tau_i) = 0$. The Heaviside function discards all solutions in the future light cone of the point, x^μ . Meanwhile, for any physical object ($v < c$), the trajectory can only cross the past light cone once. Thus, there is only one solution, $\tau_i = \tau^*$.

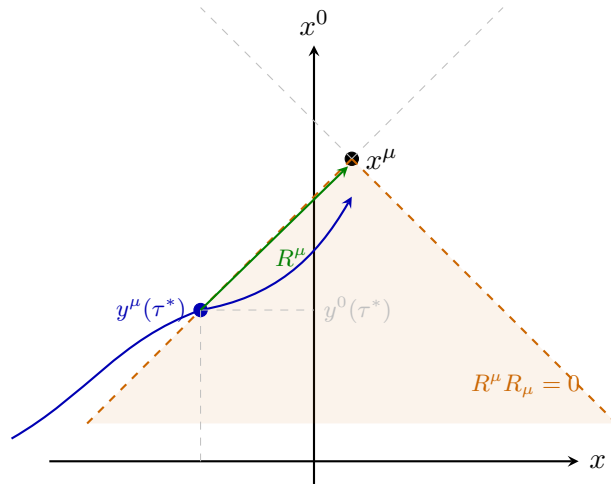


Figure 1. The worldline $y^\mu(\tau)$ (blue) has slope steeper than 45° ($v < c$), so it pierces the past light cone of x^μ exactly once at τ^* . The $\Theta(x^0 - y^0)$ factor discards intersections with the future cone. $R^\mu(\tau^*)$ is the resulting null vector.

Since $R^\nu(\tau^*)$ is light-like and $\dot{y}^\nu(\tau^*)$ is time-like, $R^\nu(\tau^*) \dot{y}_\nu(\tau^*) < 0$. Therefore, we obtain

$$A^\mu(x) = -\frac{\mu_0 qc}{4\pi} \frac{\dot{y}^\mu(\tau^*)}{R^\nu(\tau^*) \dot{y}_\nu(\tau^*)}. \quad (19)$$

Now, let us verify that the potential indeed satisfies the Lorenz gauge. First, recall that

$$R^\nu(\tau^*) R_\nu(\tau^*) = 0, \quad (20)$$

which implies

$$\partial_\mu (R^\nu(\tau^*) R_\nu(\tau^*)) = 2R^\nu(\tau^*) \partial_\mu R_\nu(\tau^*) = 2R^\nu(\tau^*) (\eta_{\mu\nu} - \dot{y}_\nu(\tau^*) \partial_\mu \tau^*) = 0. \quad (21)$$

Thus, we can write $\partial_\mu \tau^*$ in terms of R and $\dot{y}$:

$$\partial_\mu \tau^* = \frac{R_\mu(\tau^*)}{R^\nu(\tau^*) \dot{y}_\nu(\tau^*)}. \quad (22)$$

Then, consider

$$\partial_\mu A^\mu(x) \propto \partial_\mu \left(\frac{\dot{y}^\mu(\tau^*)}{R^\nu(\tau^*) \dot{y}_\nu(\tau^*)} \right) = \frac{(\partial_\mu \dot{y}^\mu) R^\nu \dot{y}_\nu - \dot{y}^\mu (\partial_\mu R^\nu) \dot{y}_\nu - \dot{y}^\mu R^\nu (\partial_\mu \dot{y}_\nu)}{(R^\rho \dot{y}_\rho)^2}, \quad (23)$$

where we have been a bit sloppy with the notations and dropped the τ^* in the second equality. The first term in the numerator can be simplified using (22):

$$(\partial_\mu \dot{y}^\mu) R^\nu \dot{y}_\nu = (\partial_\mu \tau^*) \dot{y}^\mu R^\nu \dot{y}_\nu = \dot{y}^\mu R_\mu. \quad (24)$$

Similarly, the third term becomes

$$-\dot{y}^\mu R^\nu (\partial_\mu \dot{y}_\nu) = -\dot{y}^\mu R_\mu, \quad (25)$$

which cancels the first term. We are left with the second term; using (21), we find

$$-\dot{y}^\mu (\partial_\mu R^\nu) \dot{y}_\nu = \dot{y}^\mu (\delta_\mu^\nu - \dot{y}^\nu \partial_\mu \tau^*) \dot{y}_\nu = 0, \quad (26)$$

where we have used $\dot{y}^\mu \dot{y}_\mu = -c^2$ in the last step. Since all of the terms in the numerator cancel, we have demonstrated that the potential indeed satisfies the Lorenz gauge.

To evaluate the field strength tensor, we use the similar approach as above. Consider

$$\begin{aligned} \partial^\mu A^\nu &= -\frac{\mu_0 qc}{4\pi} \frac{1}{(R^\rho \dot{y}_\rho)^2} [(\partial^\mu \dot{y}^\nu) R^\tau \dot{y}_\tau - \dot{y}^\nu (\partial^\mu R^\tau) \dot{y}_\tau - \dot{y}^\nu R^\tau (\partial^\mu \dot{y}_\tau)] \\ &= -\frac{\mu_0 qc}{4\pi} \frac{1}{(R^\rho \dot{y}_\rho)^2} \left[\ddot{y}^\nu R^\mu - \dot{y}^\nu \dot{y}^\mu - \frac{c^2 \dot{y}^\nu R^\mu}{R^\rho \dot{y}_\rho} - \frac{\dot{y}^\nu R^\tau \ddot{y}_\tau R^\mu}{R^\rho \dot{y}_\rho} \right]. \end{aligned} \quad (27)$$

Thus, we have

$$\begin{aligned} F^{\mu\nu} &= \partial^\mu A^\nu - \partial^\nu A^\mu = -\frac{\mu_0 qc}{4\pi} \frac{1}{(R^\rho \dot{y}_\rho)^2} \left[\ddot{y}^\nu - \frac{c^2 \dot{y}^\nu}{R^\rho \dot{y}_\rho} - \frac{\dot{y}^\nu R^\tau \ddot{y}_\tau}{R^\rho \dot{y}_\rho} \right] R^\mu - (\mu \leftrightarrow \nu) \\ &= -\frac{\mu_0 qc}{4\pi} \frac{1}{(R^\rho \dot{y}_\rho)^2} (R^\mu S^\nu - R^\nu S^\mu) \end{aligned} \quad (28)$$

as expected.

For a stationary charge at the origin, we have

$$y^\mu(\tau) = (c\tau, 0, 0, 0), \quad \dot{y}^\mu = (c, 0, 0, 0), \quad \text{and} \quad \ddot{y}^\mu = (0, 0, 0, 0). \quad (29)$$

Then,

$$R^\mu(\tau^*) = (x^0 - c\tau^*, \mathbf{x}). \quad (30)$$

Using $R^\mu(\tau^*) R_\mu(\tau^*) = 0$, we find

$$(x^0 - c\tau^*)^2 = |\mathbf{x}|^2, \quad (31)$$

which, in turn, gives

$$R^\mu(\tau^*) \dot{y}_\mu(\tau^*) = -c|\mathbf{x}|, \quad \text{and} \quad S^\mu = -\left(\frac{c^2}{|\mathbf{x}|}, 0, 0, 0\right). \quad (32)$$

Finally, we find

$$F^{\mu\nu} = \frac{\mu_0 qc^2}{4\pi |\mathbf{x}|^3} \begin{pmatrix} 0 & -1 & -x^2 & -x^3 \\ x^1 & 0 & 0 & 0 \\ x^2 & 0 & 0 & 0 \\ x^3 & 0 & 0 & 0 \end{pmatrix} = \frac{1}{c} \begin{pmatrix} 0 & -E^1 & -E^2 & -E^3 \\ E^1 & 0 & 0 & 0 \\ E^2 & 0 & 0 & 0 \\ E^3 & 0 & 0 & 0 \end{pmatrix} \quad (33)$$

as expected for a stationary charge at the origin since the beginning of time, where

$$\mathbf{E} = \frac{q\mathbf{x}}{4\pi\epsilon_0 |\mathbf{x}|^3}. \quad (34)$$